K Penalty Based Regulation Mechanisms

We can also design regulation mechanism $\Pi \subseteq \mathcal{C}(\mathcal{Z})$ as a contract that penalizes providers based on evidence. We assume R would be the maximum penalty and an initial credit K which is provided to cushion any penalty. Once the penalty exceeds the initial credit, the agents are then not allowed to participate in the market. We then define the notion of deterrence of the non-compliant agents as follows:

Definition K.1. (Obedience under penality) A regulation mechanism Π is said to enforce obedience under penality for a requirement $\mathfrak{R}$ if for all $P \in \Delta(\mathcal{Z})$ where $\mathfrak{R}(P) = 0$,

$$\inf_{\pi \in \Pi} \mathbb{E}_P[\pi] \geq K \tag{15}$$

The deterrence condition ensures that a non-complaint agent is unable to select a penality function that does not exceed the initial credit in expectation. Analogously we can define also feasibility as

Definition K.2. (Feasibility under penality) A regulation mechanism Π is feasible under penalty if for all $P \in \Delta(\mathcal{Z})$ such that $\mathfrak{R}(P) = 1$ there exists a $\pi \in \Pi$ such that $K > \mathbb{E}_P[\pi]$.

Now the Def. 3.4 for implementability of a requirement via a regulation mechanism applies directly where we call a penalty based regulation mechanism to implement a requirement $\mathfrak{R}$ if non-compliant agents choose to self exclude while compliant agents have incentive to participate with $G(\Pi, P) = \mathbb{1}[\inf_{\pi \in \Pi} \mathbb{E}_P[\pi(Z)] < K]$.

Corollary K.3 (Penalty characterisation)**.** *Let $\mathfrak{R}$ be a requirement with $\mathcal{P}_0 = \{P : \mathfrak{R}(P) = 0\}$. Then there exists a penalty mechanism (Π, K) that implements $\mathfrak{R}$* if and only if *$\mathcal{P}_0$ is a credal set.*

Proof. Let $T : \mathcal{C}(\mathcal{Z}; [0, R]) \to \mathcal{C}(\mathcal{Z}; [0, R])$, $T(\pi) = R - \pi$; T is an affine bijection, and $\mathbb{E}_P[T(\pi)] = R - \mathbb{E}_P[\pi]$. Which yields the following transformation identities

$$\inf_{\pi \in M} \mathbb{E}_P[\pi] \geq K \iff \sup_{\pi' \in T(M)} \mathbb{E}_P[\pi'] \leq R - K \tag{16}$$

$$\big(\exists\, \pi \in M : \mathbb{E}_P[\pi] < K\big) \iff \big(\exists\, \pi' \in T(M) : \mathbb{E}_P[\pi'] > R - K\big). \tag{17}$$

$(\Leftarrow)$

We show that if $\mathcal{P}_0$ is a credal set then a penalty mechanism Π with credit K implements $\mathfrak{R}$. Since $\mathcal{P}_0$ is a credal set, based on theorem 3.5 there exists a license mechanism $\Lambda \subseteq \mathcal{C}()$ with entry fee C that implements $\mathfrak{R}$. Put $\Pi := T(\Lambda)$. Applying Eq 16 on $\{P : \mathfrak{R}(P) = 0\}$ and Eq 17 on $\{P : \mathfrak{R}(P) = 1\}$ shows that there exists a (Π, K) with $K = R - C$that satisfies obedience and feasibility under-penalty.

$(\Rightarrow)$

We now show if a penalty mechanism implements $\mathfrak{R}$ then $\mathcal{P}_0$ credal. Suppose (Π, K) achieves PMO under penalty, and assume for contradiction that $\mathcal{P}_0$ is *not* credal. Let $\Lambda := T(\Pi)$ with entry fee $R - K$. Reading conditions in Eq 16 and Eq 17 right-to-left, Λ satisfies obedience (Def 3.2) and feasibility (Def 3.3). Thus, there exists a license based regulation mechanism that achieves PMO for the requirement whose non-compliant set is $\mathcal{P}_0$. By Theorem 3.5 this forces $\mathcal{P}_0$ to be a credal set contradicting the assumption. Hence $\mathcal{P}_0$ is credal. ☐